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**C2****Started on** Thursday, 17 September 2026, 8:24 AM**State** Finished**Completed on** Thursday, 17 September 2026, 8:24 AM**Time taken** 33 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

**Question 1** | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

|    |          |   |          |          |          |          |          |
|----|----------|---|----------|----------|----------|----------|----------|
| s1 | a        | g | <b>g</b> | <b>t</b> | <b>a</b> | <b>b</b> |          |
| s2 | <b>g</b> | x | <b>t</b> | x        | <b>a</b> | <b>y</b> | <b>b</b> |

**The length is 4**

Solving it using Dynamic Programming

For example:

| Input | Result |
|-------|--------|
| aab   | 2      |
| azb   |        |

**Answer:** (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <string.h>
3
4  #define MAX_LEN 1005
5
6  int dp[MAX_LEN][MAX_LEN];
7
8  int max(int a, int b) {
9      return (a > b) ? a : b;
10 }
11
12 int main() {
13     char s1[MAX_LEN];
14     char s2[MAX_LEN];
15
16     // Read the two strings silently from standard input
17     if (scanf("%s", s1) != 1) return 0;
18     if (scanf("%s", s2) != 1) return 0;
19
20     int m = strlen(s1);
21     int n = strlen(s2);
22
23     // Build the DP table from the bottom up
24     for (int i = 0; i <= m; i++) {
25         for (int j = 0; j <= n; j++) {
26             if (i == 0 || j == 0) {
27                 dp[i][j] = 0; // Base case: empty strings have 0 common subsequence length
28             } else if (s1[i - 1] == s2[j - 1]) {
29                 dp[i][j] = dp[i - 1][j - 1] + 1;
30             } else {
31                 dp[i][j] = max(dp[i - 1][j], dp[i][j - 1]);
32             }
33         }
34     }
35
36     // The bottom right cell contains the length of the longest common subsequence

```

```
36 // the bottom-right cell contains the length of the longest common subsequence
37 printf("%d\n", dp[m][n]);
38
39 return 0;
40 }
41
```

|   | Input        | Expected | Got |   |
|---|--------------|----------|-----|---|
| ✓ | aab<br>azb   | 2        | 2   | ✓ |
| ✓ | ABCD<br>ABCD | 4        | 4   | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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